EXPERIMENT 6

Automated Business Workflow using Deluge Scripting

Application Testing and Debugging — EcoCycle E-Waste Entry (Priority Calculation Use Case)

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1. Aim

To design and implement an automated business-process workflow in the EcoCycle – E-Waste Collection & Recycling Management application using Deluge scripting, so that every submitted E-Waste Entry gets a priority score, a priority level (High / Medium / Low) and its initial statuses automatically, and to test and debug the result on the live application.

Objectives

- To configure a form workflow that runs on the successful creation of an E-Waste Entry record.
- To write Deluge logic that reads quantity, age, hazardous flag and condition and calculates the Priority Score.
- To classify the score as High (8 or above), Medium (4 to 7) or Low (0 to 3) and set Collection Status = Pending and Recycling Status = Not Processed.
- To add a conditional e-mail alert for High-priority e-waste and a past-date check on the Collection Request form.
- To verify the behaviour through the reports, the dashboard and a set of test entries.

2. Procedure

Step 1: Creating the workflow

1. Open the EcoCycle application in Edit mode and select **Workflow** from the top navigation.
2. Click **Create New Workflow** and choose the form **E-Waste Entry**.
3. Set the Record Event to **Created** and the Form Event to **Successful form submission**.
4. Name the workflow **Calculate Priority** and click Create Workflow.

Setting	Value Selected	Purpose
Choose form	E-Waste Entry	Links the workflow to the form where items are recorded.
Record Event	Created	Runs only for a new item, so the score is not recalculated on every edit and e-mails are not repeated.
Form Event	Successful form submission	Runs after the mandatory fields (name, category, quantity, age, hazardous flag, location, user name, e-mail) have passed validation and the record is saved.
Workflow name	Calculate Priority	Identifies the purpose of the workflow.

Step 2: Adding the Deluge script action

1. Inside the workflow, click **Add New Action** and choose **Deluge script** from the Action type grid.
2. Enter the script of Listing 6.1. It adds points for quantity, age, hazardous material and condition, compares the total with 8 and 4, and writes the score, level and the two initial statuses back to the record.

3. Click **Update** so that Creator validates the syntax and saves the script.

Listing 6.1: Priority calculation and status defaults (DS-01)

```
// Form: E-Waste Entry | Event: Created > Successful form submission
score = 0;
// Quantity
if(input.Quantity >= 8)       { score = score + 3; }
else if(input.Quantity >= 4)  { score = score + 2; }
else if(input.Quantity >= 2)  { score = score + 1; }
// Age in years
if(input.Age_years >= 3)      { score = score + 2; }
else if(input.Age_years >= 1) { score = score + 1; }
// Hazardous material
if(input.Contains_Hazardous_Material == "Yes") { score = score + 3; }
// Condition
if(input.Condition == "Not Working" || input.Condition == "Damaged")
{
    score = score + 2;
}
// Level from the score ranges
if(score >= 8)      { level = "High"; }
else if(score >= 4) { level = "Medium"; }
else                { level = "Low"; }
rec = E_Waste_Entry[ID == input.ID];
rec.Priority_Score    = score;
rec.Priority_Level    = level;
rec.Collection_Status = "Pending";
rec.Recycling_Status  = "Not Processed";
```

Note: the score ranges (8+, 4–7, 0–3) are those of the project. The factor weights and field link names in this listing are the reference scheme used in the project report (Appendix C); replace them with the application's own values if they differ.

Step 3: Adding the High-priority e-mail alert

1. Add a second action in the same workflow (after the score is set) and choose **Run this action: on condition**.
2. Set the condition to **Priority Level equals High**.
3. Add a Deluge **sendmail** block that mentions the item name, quantity, collection location and score (Listing 6.2).

Listing 6.2: High-priority e-mail (DS-03)

```
if(input.Priority_Level == "High")
{
    sendmail
    [
        from : zoho.adminuserid
        to : zoho.adminuserid
        subject : "High-priority e-waste: " + input.E_Waste_Name
        message : "Quantity " + input.Quantity + ", collection location "
                  + input.Collection_Location + ", priority score "
                  + input.Priority_Score + "."
    ]
}
```

Step 4: Past-date validation on Collection Request

1. Open the Collection Request form and add a workflow with the Form Event **On validate**.
2. Add the script of Listing 6.3 so that a preferred collection date earlier than today is rejected.

Listing 6.3: Past-date check (DS-02)

```
if(input.Preferred_Collection_Date < zoho.currentdate)
{
    alert "The preferred collection date cannot be in the past.";
    cancel submit;
}
```

Step 5: Live testing

1. Switch to the live application and open the **E-Waste Entry** form (Figures 6.1 and 6.2).
2. Submit the four test items of Table 6.1 and open **All E-Waste** and **High Priority E-Waste**.
3. Open the **EcoCycle Dashboard** and compare the pie chart with the expected split.

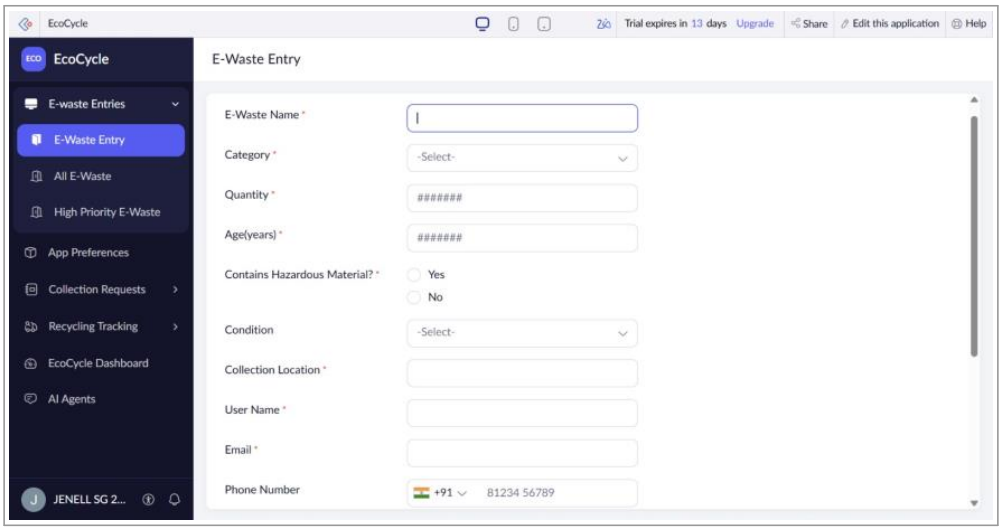


Figure 6.1: E-Waste Entry form (upper part) — the inputs read by the workflow

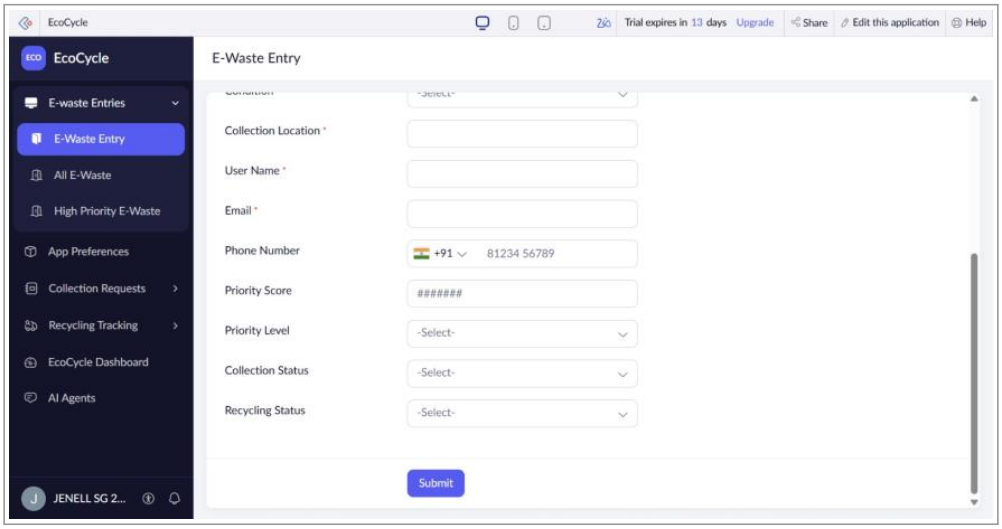


Figure 6.2: E-Waste Entry form (lower part) — Priority Score, Priority Level and the two statuses are filled by the workflow

3. Test Data and Expected Results

Table 6.1 lists the four entries stored in the application. The last two columns apply the reference scheme of Listing 6.1; the stored scores themselves are not visible in the report screenshots, so the scores below are expected values.

Entry	Qty	Age (yrs)	Hazardous	Condition	Expected Score	Expected Level
laptop	5	3	No	Working	4	Medium
computer	2	1	No	Working	2	Low
Printer	8	5	Yes	Not Working	10	High
monitor	10	5	Yes	Damaged	10	High

4. Output & System Verification

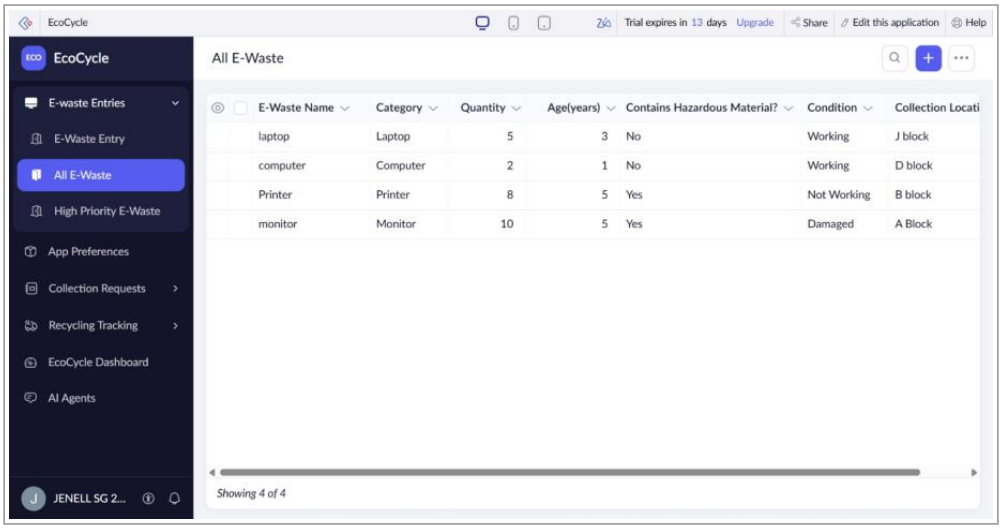


Figure 6.3: All E-Waste report — four records (Showing 4 of 4)

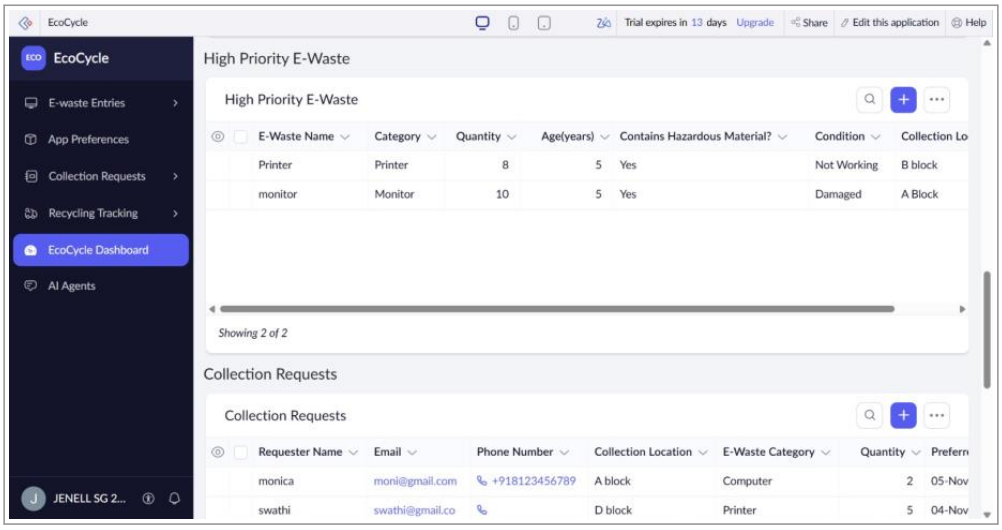


Figure 6.4: High Priority E-Waste report on the dashboard — Printer and monitor only (Showing 2 of 2)

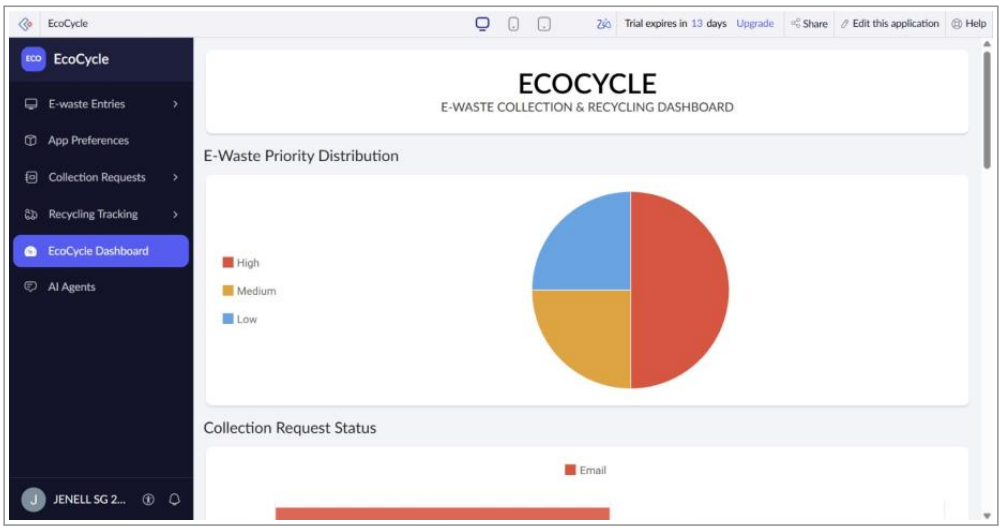


Figure 6.5: E-Waste Priority Distribution pie chart — High 50%, Medium 25%, Low 25%

Observations

- Four entries are stored and all of them appear in All E-Waste (Figure 6.3).
- Only the Printer (quantity 8, hazardous, Not Working) and the monitor (quantity 10, hazardous, Damaged) appear in High Priority E-Waste (Figure 6.4).
- The pie chart shows half of the entries as High and a quarter each as Medium and Low (Figure 6.5), which agrees with the expected levels in Table 6.1 (two High, one Medium, one Low).
- The recycling status of new records is Not Processed (Recycling Status chart, Figure 5.11 of the project report).

Debugging notes and pending tests

- Deluge's **info** statement can be added after the score is calculated to write the score to the log while testing.
- A blank or negative quantity or age would distort the score; a field validation on E-Waste Entry is recommended.
- Boundary entries scoring exactly 3, 4, 7 and 8, the e-mail alert and the past-date rejection were not captured in the project evidence and are listed as planned tests (TC-08, TC-09 to TC-16 of the project report).

5. Result

A form workflow written in Deluge was configured on the E-Waste Entry form. On submission it calculates the priority score, classifies the item as High, Medium or Low, and initialises the collection and recycling statuses. The levels shown in the High Priority E-Waste report and the dashboard pie chart agree with the expected levels for the four test entries.

6. Conclusion

The experiment shows how a short Deluge script replaces manual judgement of urgency with a rule that is applied the same way for every item. Choosing *Created* and *Successful form submission* as the triggers keeps the score from being recalculated on edits and ensures it uses validated data. Documenting the factor weights and running the boundary tests are the next steps to make the workflow fully auditable.